which orbit around stellar companions, a few (though not most) might shine brightly with X-rays.

To search for black holes, then, one could use a combination of optical telescopes and X-ray telescopes. The black-hole candidates would be binaries in which one object is an optically bright but X-ray-dark star, and the other is an optically dark but X-ray-bright object (the black hole). Since a neutron star could also capture gas from a companion, heat it in shock fronts, and produce X-rays, the weighing of the optically dark but X-ray-bright object would be crucial. One must be sure it is heavier than 2 Suns and thus not a neutron star.

There was but one problem with this search strategy. In 1966, X-ray telescopes were extremely primitive.

The Search

The trouble with X-rays, if you are an astronomer, is that they cannot penetrate the Earth's atmosphere. (If you are a human, that is a virtue, since X-rays cause cancer and mutations.)

Fortunately, experimental physicists with vision, led by Herbert Friedman of the D.S. Naval Research Laboratory (NRL), had been working since the 1940s to lay the groundwork for space-based X-ray astronomy. Friedman and his colleagues had begun, soon after World War 11, by flying instruments to study the Sun on captured German V-2 rockets. Friedman has described their first flight, on 28 June 1946, which carried in the rocket's nose a spectrograph for studying the Sun's far ultraviolet radiation. (Far ultraviolet rays, like X-rays, cannot penetrate the Earth's atmosphere.) After soaring above the atmosphere briefly and collecting data, "the rocket returned to Earth, nose down, in streamlined flight and buried itself in an enormous crater some 80 feet in diameter and 30 feet deep. Several weeks of digging recovered just a small heap of unidentifiable debris; it was as if the rocket had vaporized on impact."

From this inauspicious beginning, the inventiveness, persistence, and hard work of Friedman and others brought ultraviolet and X-ray astronomy step by step to fruition. By 1949 Friedman and his colleagues were flying Geiger counters on V-2 rockets to study X-rays from the Sun. By the late 1950s, now flying their counters on American-made Aerobee rockets,